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Splichal

(54) **DUAL SENSORY BALANCE BOARD AND METHODS OF USE THEREOF**

(71) Applicant: **Naboso Technology, Inc.**, Chandler, AZ (US)

(72) Inventor: **Emily Splichal**, Chandler, AZ (US)

(73) Assignee: **Naboso Technology, Inc.**, Chandler, AZ (US)

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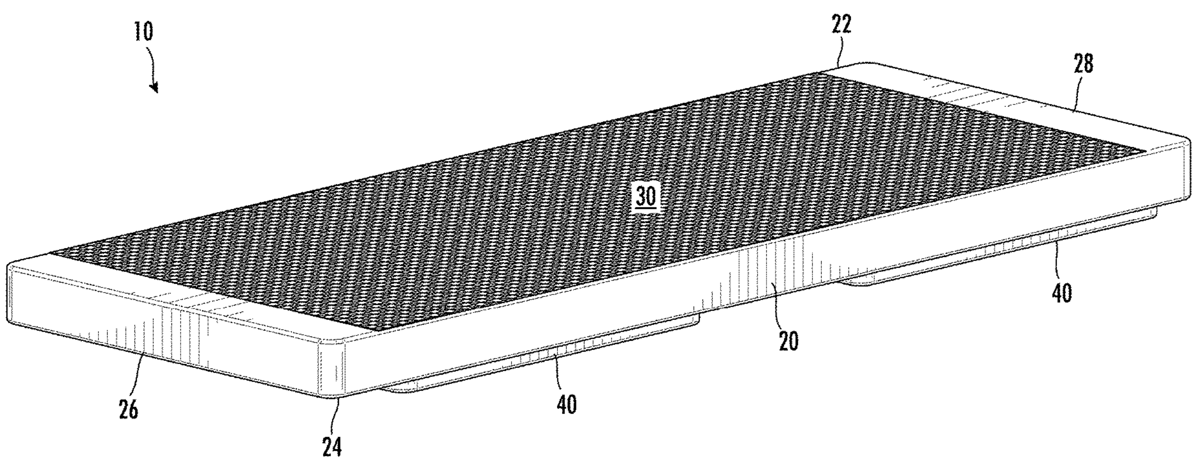
(57) **ABSTRACT**

A balance board including a platform and a first balance block. The platform having a top side and a bottom side that is opposite the top side. The platform is sized and dimensioned to receive a single foot of a user thereon. The top side of the platform may include a mechanoreception stimulation surface that is configured to mechanoreceptively stimulate neurons within the cutaneous skin of the user. The first balance block is removably attached to the bottom side of the platform. The first balance block that is configured to proprioceptively stimulate neurons within muscles, tendons, or joints of the user.



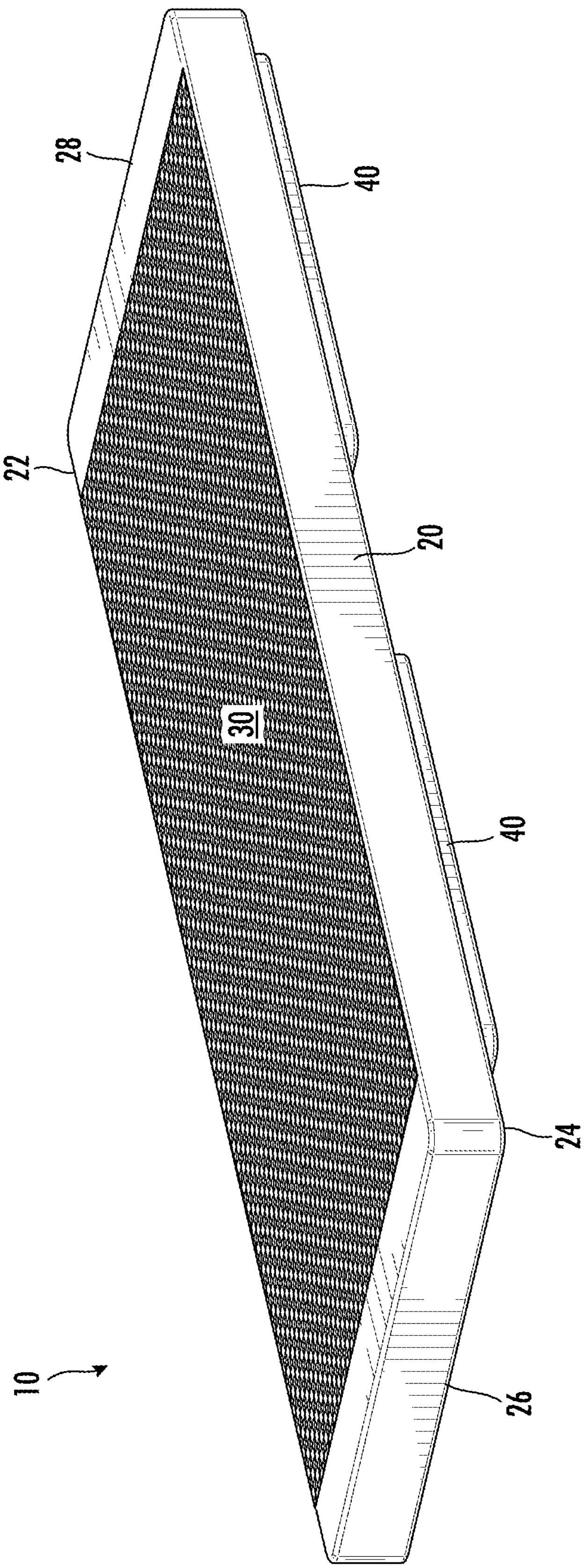


FIG. 1

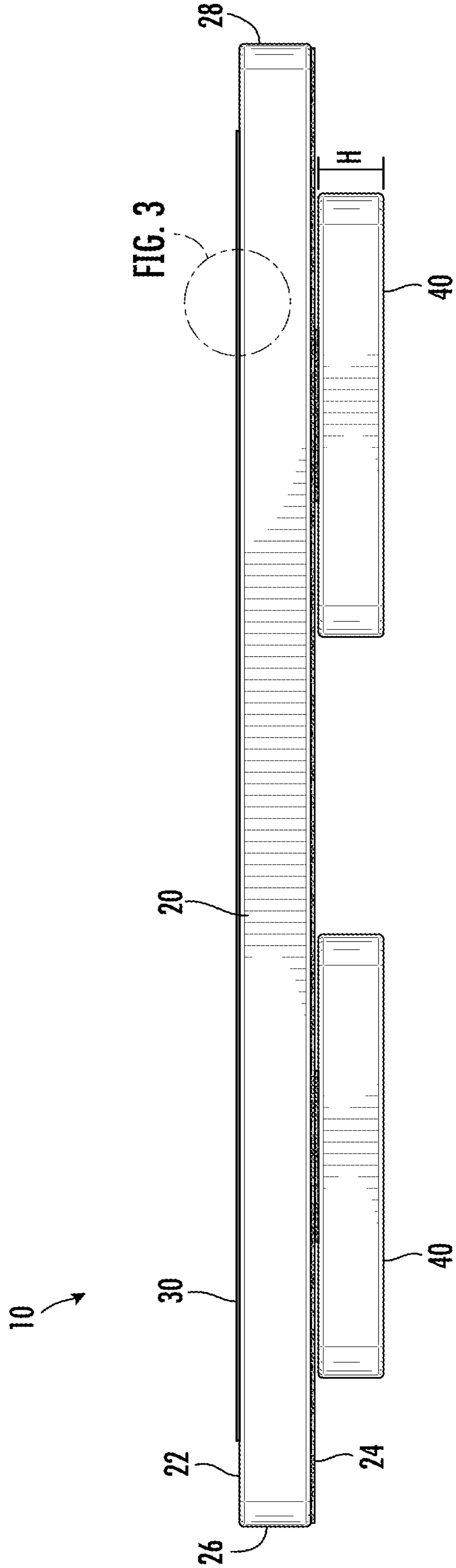


FIG. 2

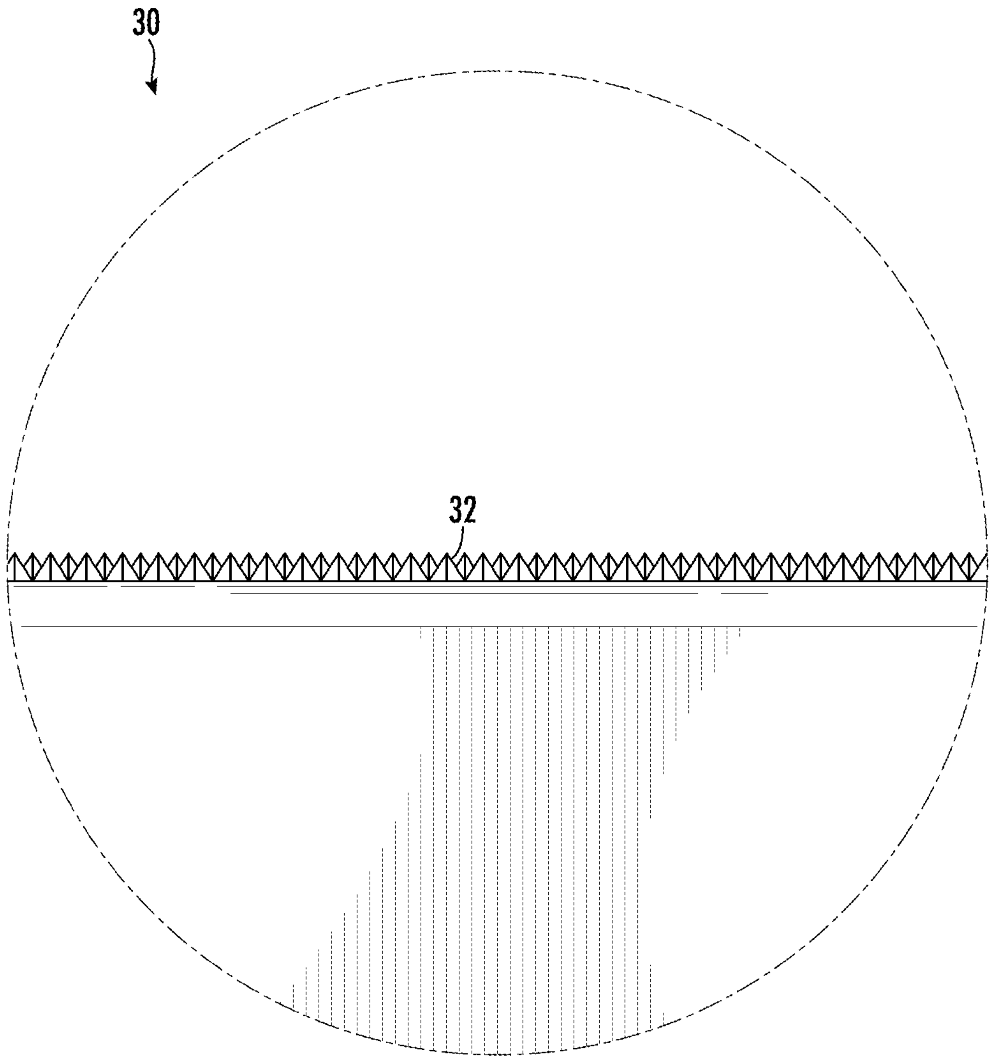


FIG. 3

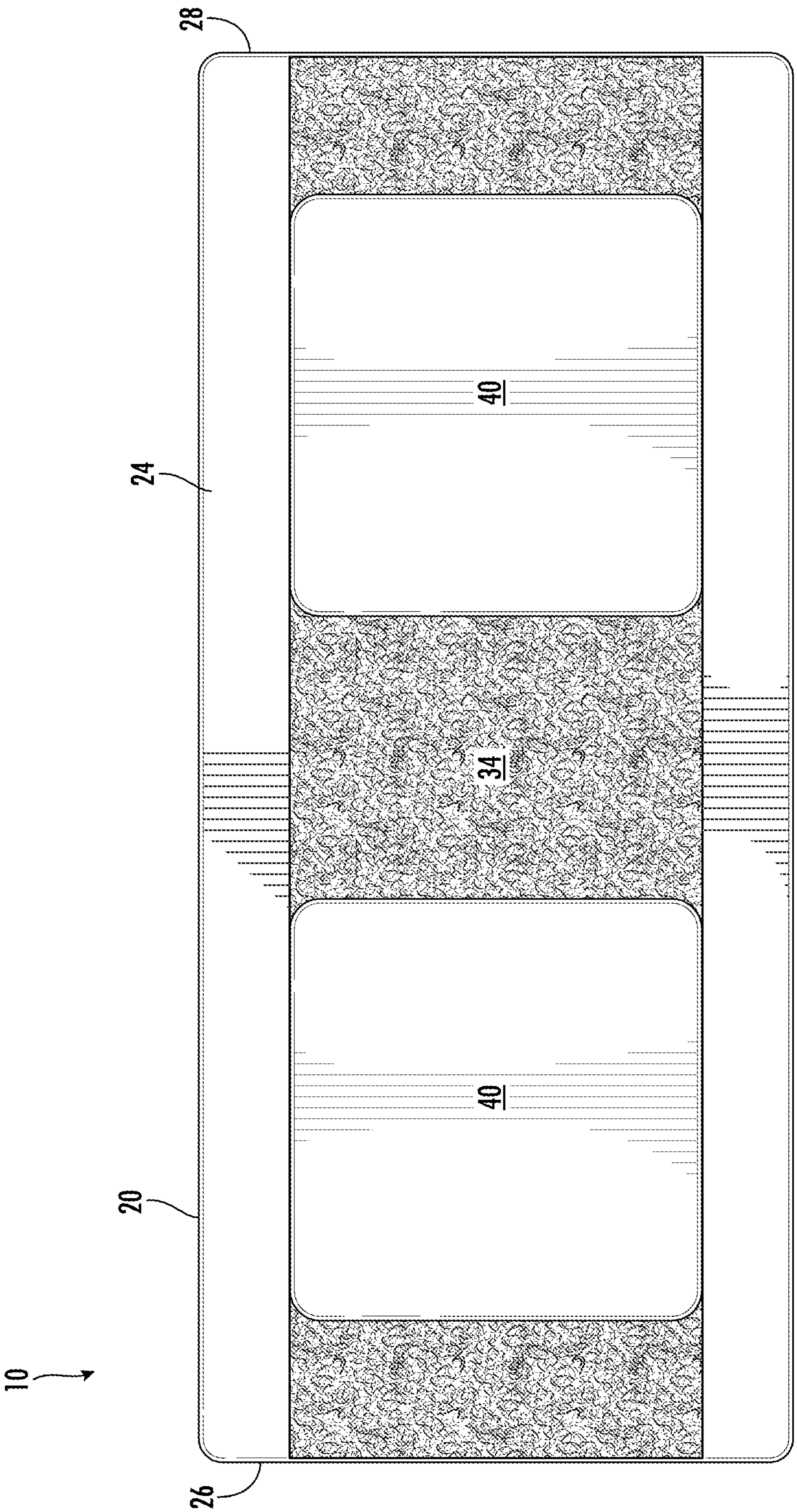


FIG. 4

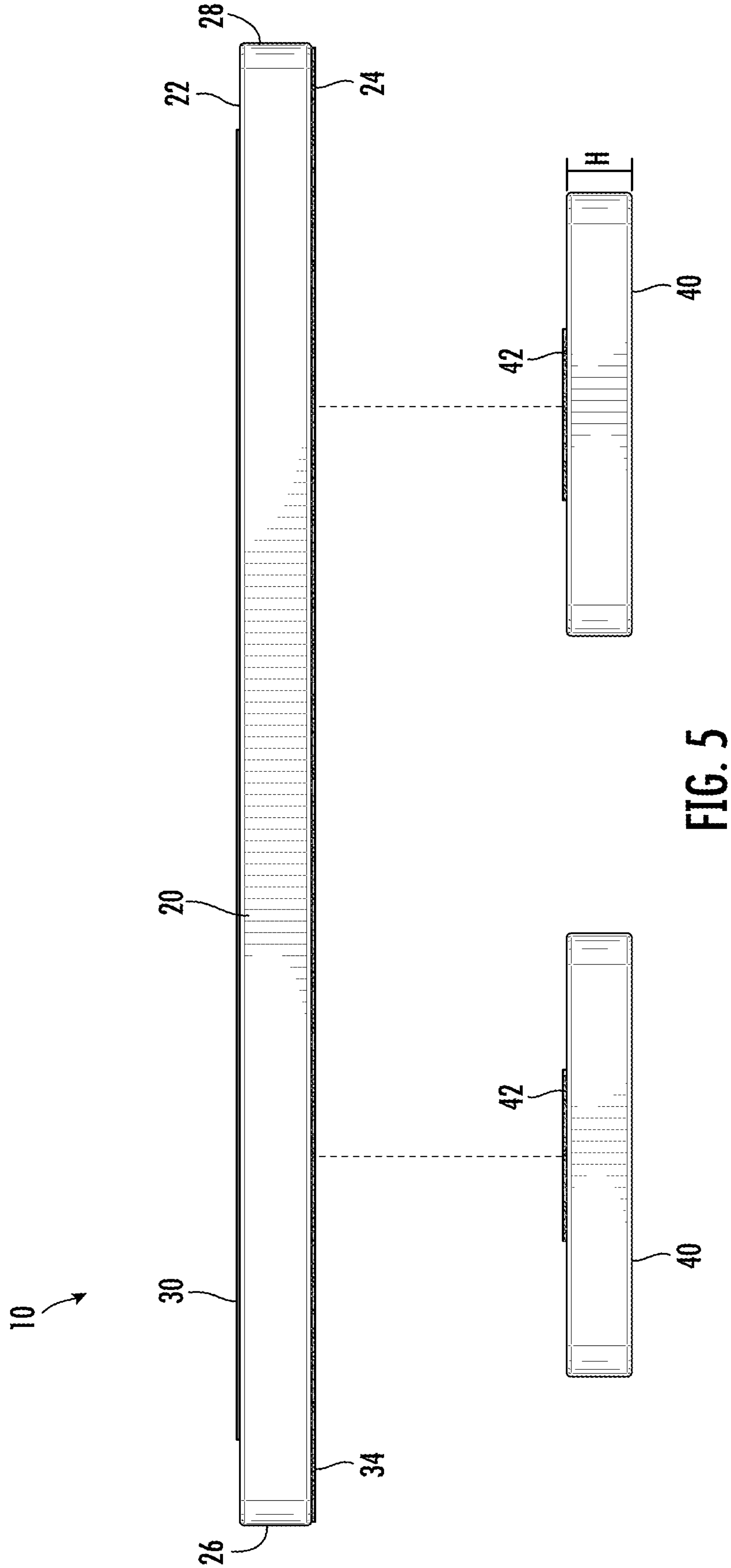


FIG. 5

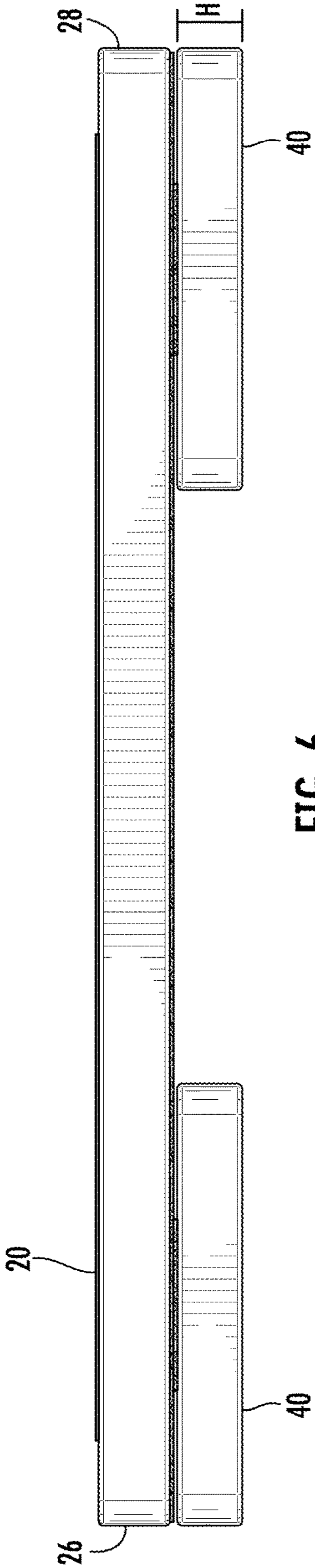


FIG. 6

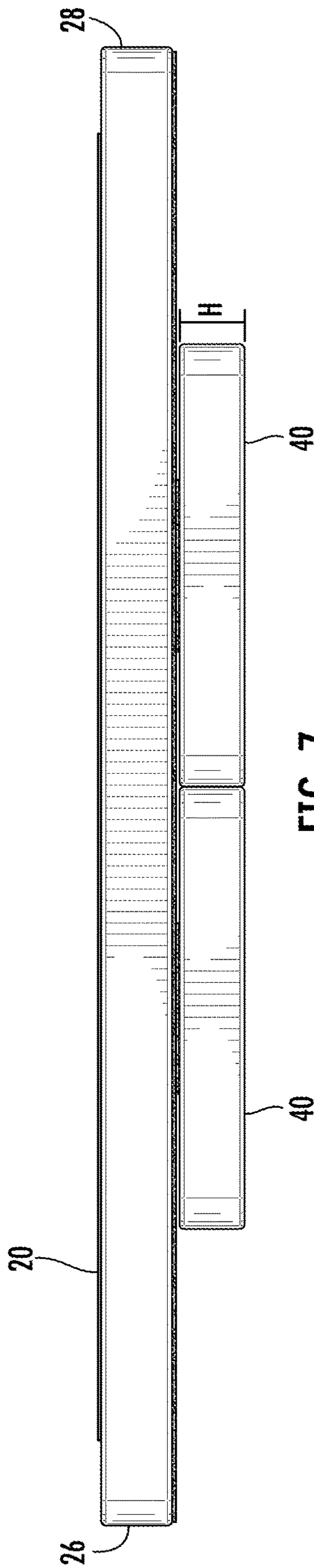


FIG. 7

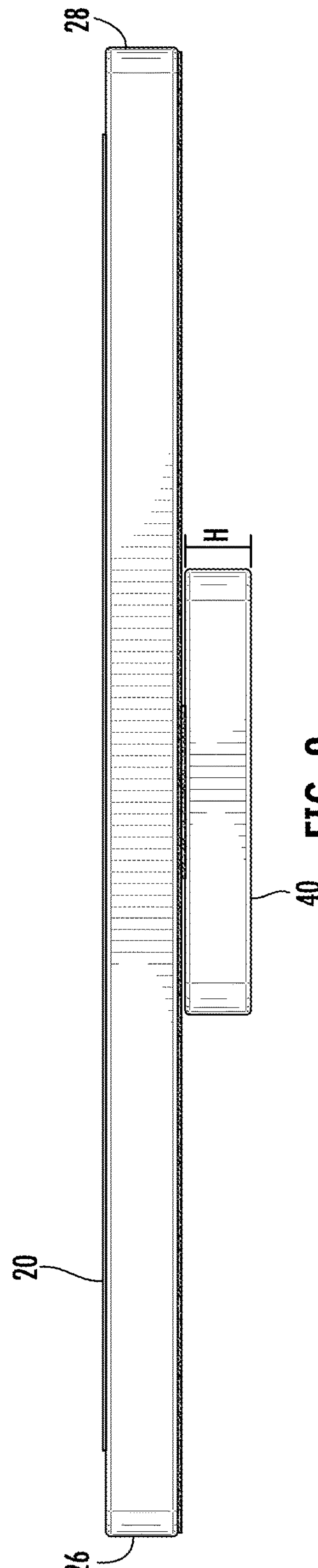


FIG. 8

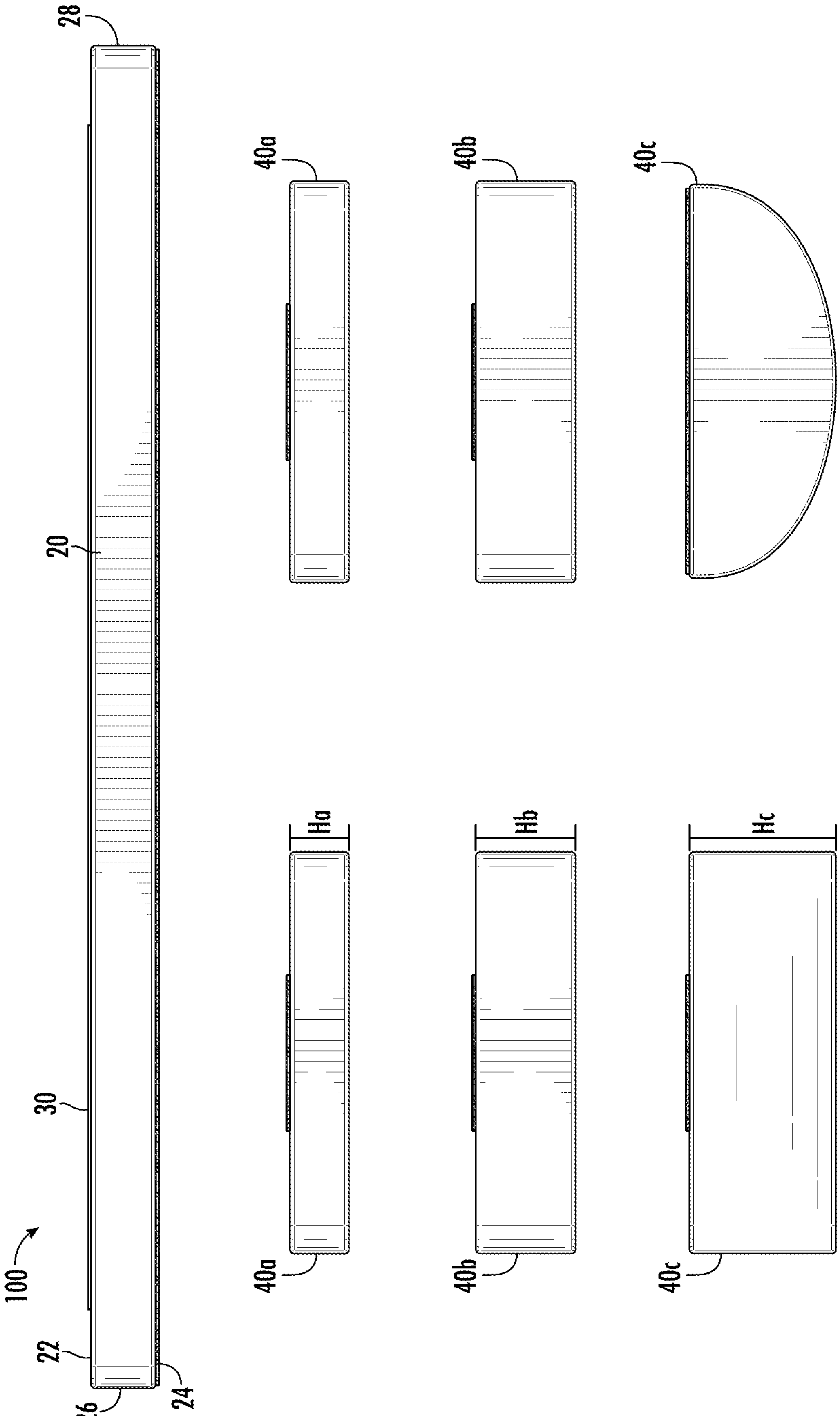


FIG. 9

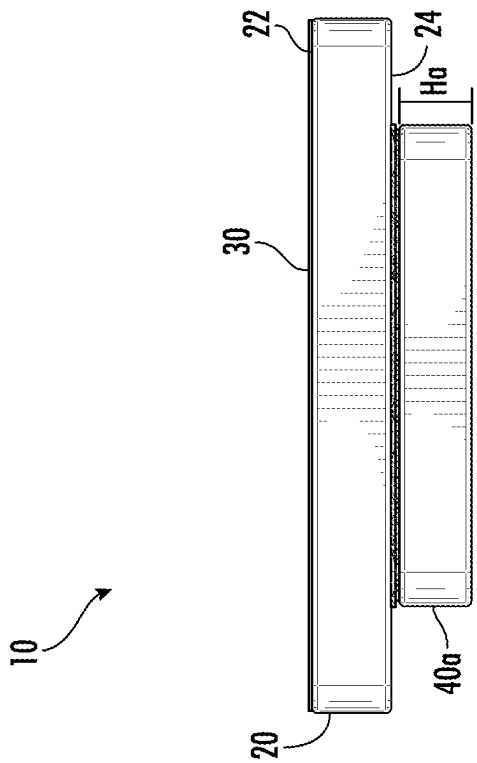


FIG. 10

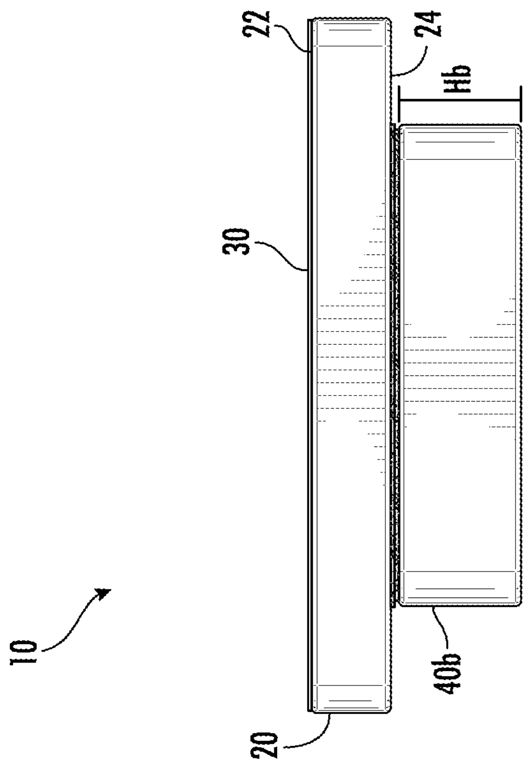


FIG. 11

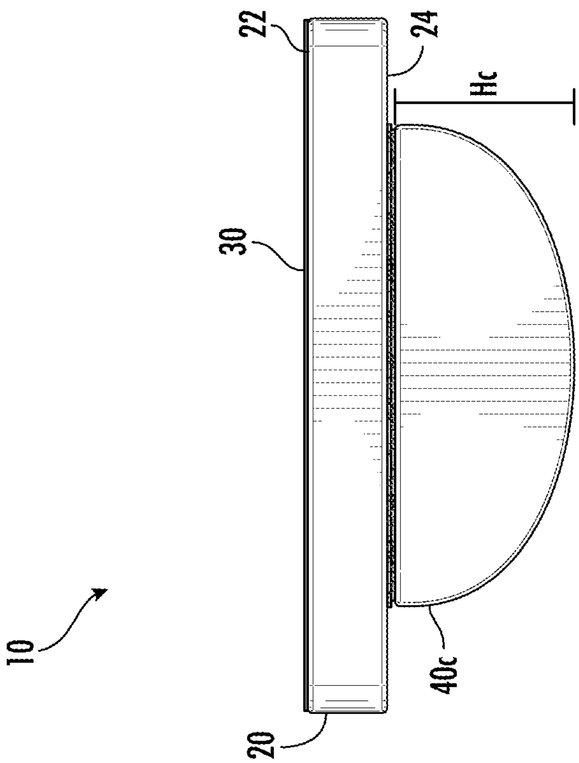


FIG. 12

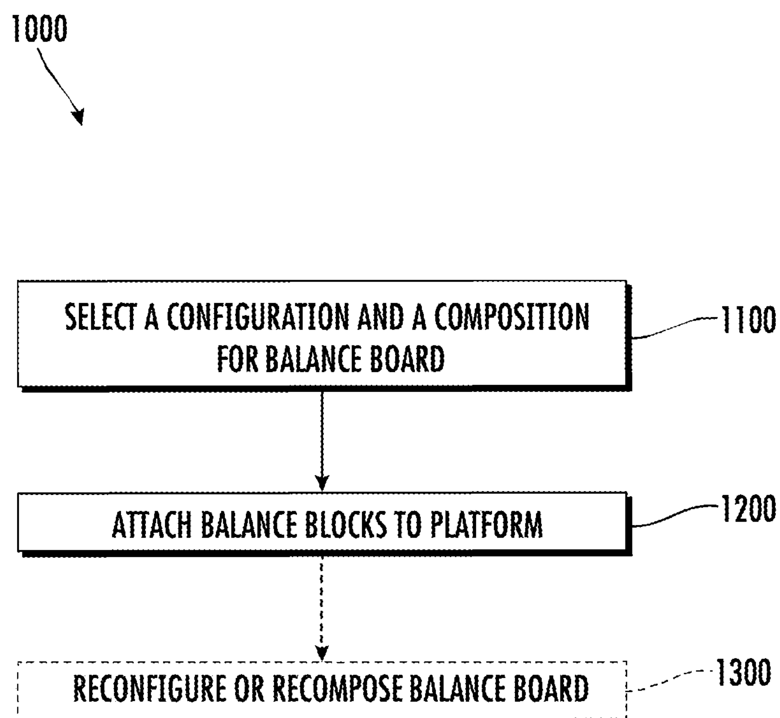

1000
SELECT A CONFIGURATION AND A COMPOSITION FOR BALANCE BOARD
1100
ATTACH BALANCE BLOCKS TO PLATFORM
1200
RECONFIGURE OR RECOMPOSE BALANCE BOARD
1300

FIG. 13

DUAL SENSORY BALANCE BOARD AND METHODS OF USE THEREOF

BACKGROUND

1. Technical Field

[0001] The present disclosure relates to fitness equipment and therapeutic devices and, more specifically, to balance or wobble boards to train and improve balance and locomotion.

2. Discussion of Related Art

[0002] Balance and ambulation are a complex relation between several biological systems such as the central nervous system (CNS), peripheral nervous system (PNS), and the skeletal muscle system. One aspect between these systems is proprioception, or the sense of self-movement. Proprioceptors in the muscles, tendons, and joints detect kinematic information about the limb, joint, or digit such as joint position, movement, or load. The proprioceptors share this kinematic information with the CNS where it is integrated with information from other sensory systems, e.g., the visual system and the vestibular system, to create a wholistic representation of body position, movement, and acceleration.

[0003] Balance or wobble boards have been used by physical therapists and athletes to improve balance and locomotion for both rehabilitative and performance purposes. Balance boards challenge the relationships between the central nervous system (CNS), peripheral nervous system (PNS), and the skeletal muscle system to train and improve balance and ambulation. Conventional balance boards often include a support surface and a rocking or wobbling base. A user stands on the support surface and the base is configured to be intentionally unstable beneath the user, forcing the user to actively balance on the balance board.

[0004] Separate devices may improve balance through mechanical stimuli to the bottoms of the feet. The soles of the feet contain mechanoreceptors within the cutaneous layer of skin that sense mechanical stimuli such as pressure, skin stretch, or texture. Such mechanical stimuli may increase local nerve activation and blood flow. Example devices for providing mechanical stimuli are described in U.S. Pat. No. 11,642,279, filed Sep. 22, 2020, and U.S. Patent Publication No. 20220134046, filed Jan. 14, 2022.

SUMMARY

[0005] This disclosure relates generally to a balance board that simultaneously stimulates both mechanoreceptors and proprioceptors of a user. The dual stimulation may enhance somatosensory stimulation and may further benefit balance training.

[0006] In an aspect of the present disclosure, a balance board includes a platform and first balance block. The platform has a top side and a bottom side opposite the top side. The platform is sized and dimensioned to receive a single foot of a user thereon. The top side of the platform includes a mechanoreception stimulation surface configured to mechanoreceptively stimulate neurons within cutaneous skin of the user. The first balance block is removably attached to the bottom side of the platform. The first balance block is configured to proprioceptive stimulate neurons within muscles, tendons, or joints of the user.

[0007] In aspects, the top side is configured to receive the single foot of the user in an unsecured manner. The mechanoreception stimulation surface may include a plurality of protuberances projecting away from the top side of the platform.

[0008] In some aspects, the first balance block is configured to micro-wobble under the weight of a user to proprioceptively stimulate neurons of the user. The first balance block may be configured to micro-wobble omnidirectionally around a perimeter thereof to improve balance of the user in each plane or axis of movement.

[0009] In certain aspects, the balance board includes a second balance block removably attached to the bottom side of the platform. The second balance block may be sized and dimensioned similar to the first balance block. The balance board may have a first configuration in which the first balance block is disposed adjacent to a first side end of the platform and the second balance block is disposed adjacent to a second side end of the platform opposite the first side end. The second balance block may be spaced apart from the first balance block. The balance board may have a second configuration in which the first balance block is spaced apart from the first side end and the second balance block is spaced apart from the second side end. The balance board may be configured to provide more proprioceptive stimulate when in the second configuration than in the first configuration.

[0010] In another aspect of the present disclosure, a balance board includes a platform and a first balance block. The platform has a textured first side and second side opposite the textured first side. The second side includes a first part of a two-part fastener. The platform is sized and dimensioned to receive a single foot of a user thereon, the textured first side is configured to mechanoreceptivley stimulate the user. The first balance block includes a second part of the two-part fastener. The first balance block is removably attached to the second side of the platform. The first balance block is configured to proprioceptively stimulate the user. The balance board is configured to improve balance of the user during locomotion.

[0011] In aspects, the textured first side comprises a plurality of protuberances projecting upwardly therefrom. The first balance block may be configured to deform under the weight of the user to proprioceptively stimulate neurons of the user. The first balance block may be configured to deform omnidirectionally around a perimeter thereof under the weight of the user to improve balance of the user in each plane or axis of movement. The balance board may be configured to improve balance of the user with the single foot of the user unsecured to the platform.

[0012] In some aspects, a balance trainer kit includes a balance board with a plurality of balance blocks. The balance blocks are attachable to the platform such that the balance board is configurable into a first configuration, a second configuration, and a third configuration and is composable into a first composition and a second composition.

[0013] In another aspect of the present disclosure, a balance board includes a platform having a top side and a bottom side opposite the top side. The top side includes a mechanoreception stimulation surface. The bottom side including a securement surface extending along a length of the bottom side between a first side end of the platform and a second side end opposite the first side end. The first balance block is removably attached to the bottom side of the platform and is positionable between the first side end and the second side end. The first balance block includes an attachment surface mateable with the securement surface to attach the first balance block to the platform.

[0014] In aspects, the balance block has a rectangular profile. The first balance block may have a height in the range of 10 millimeters to 60 millimeters. The first balance block may have a semi-circular profile. The securement surface may be a first part of a two-part fastener and the attachment surface may be a second part of the two-part fastener.

[0015] In some aspects, the balance board includes a second balance block removably attached to the bottom side of the platform and positionable between the first side end and the second side end. The balance board may have a first configuration in which the first balance block is disposed adjacent to the first side end and the second balance block is disposed adjacent to the second side end and spaced apart from the first balance block. The balance board may have a second configuration in which the first balance block is spaced apart from the first side end and the second balance block is spaced apart from the second side end.

[0016] In another aspect of the present disclosure, a method of assembling a balance board includes selecting a configuration and a composition for the balance board and attaching balance blocks to the platform. A configuration for the balance board is selected from one of a first configuration, a second configuration, or a third configuration. A composition for the balance board is selected from one of a first composition, a second composition, or a third composition. The balance blocks are attached to a platform of the balance board such that the balance board is in the selected configuration and has the selected composition.

[0017] In aspects, the method includes reconfiguring the balance board into a different configuration of the first configuration, the second configuration, or the third configuration. The method may include recomposing the balance board into a different composition of the first composition, the second composition, or the third composition.

[0018] Further, to the extent consistent, any of the embodiments or aspects described herein may be used in conjunction with any or all of the other embodiments or aspects described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Various aspects of the present disclosure are described hereinbelow with reference to the drawings, which are not necessarily drawn to scale, which are incorporated in and constitute a part of this specification, wherein:

[0020] FIG. **1** is a perspective view of a balance board in accordance with embodiments of the present disclosure;

[0021] FIG. **2** is a side view of the balance board of FIG. **1**;

[0022] FIG. **3** is an enlarged view of a portion of the balance board as indicated in FIG. **2**;

[0023] FIG. **4** is a bottom view of the balance board of FIG. **1**;

[0024] FIG. **5** is an exploded view of the balance board of FIG. **1**;

[0025] FIG. **6** is a side view of the balance board of FIG. **1** in a first configuration thereof;

[0026] FIG. **7** is a side view of the balance board of FIG. **1** in a second configuration thereof;

[0027] FIG. **8** is a side view of the balance board of FIG. **1** in a third configuration thereof;

[0028] FIG. **9** is a side view of a balance trainer kit in accordance with embodiments of the present disclosure;

[0029] FIG. **10** is an end view of a balance board in accordance with embodiments of the present disclosure in a first composition thereof;

[0030] FIG. **11** is an end view of a balance board in accordance with embodiments of the present disclosure in a second composition thereof;

[0031] FIG. **12** is an end view of a balance board in accordance with embodiments of the present disclosure in a third composition thereof; and

[0032] FIG. **13** is a flowchart illustrating a method of assembling a balance board in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0033] The present disclosure will now be described more fully hereinafter with reference to example embodiments thereof with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. These example embodiments are described so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Features from one embodiment or aspect can be combined with features from any other embodiment or aspect in any appropriate combination. For example, any individual or collective features of method aspects or embodiments can be applied to apparatus, product, or component aspects or embodiments and vice versa. The disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. As used in the specification and the appended claims, the singular forms "a," "an," "the," and the like include plural referents unless the context clearly dictates otherwise. In addition, while reference may be made herein to quantitative measures, values, geometric relationships or the like, unless otherwise stated, any one or more if not all of these may be absolute or approximate to account for acceptable variations that may occur, such as those due to manufacturing or engineering tolerances or the like.

[0034] As used herein, the term "locomotion" refers to natural or functional motion of humans such as walking or running. The term "proprioceptive stimulation" refers to activation of neurons within the muscles, tendons, ligaments, or joints of a user that sense kinematic parameters of a joint such as position, movement, or load. The term "mechanoreceptive stimulation" refers to activation of neurons within the dermis layer of skin of a user that sense mechanical pressure. The term "macro-circulation" refers to circulation of blood through the relatively large blood vessels of the circulatory system such as arteries and veins and the term "micro-circulation" refers to circulation of blood through the smallest blood vessels of the circulatory system such as arterioles, capillaries, or venules.

[0035] Referring now to FIG. **1**, a balance board **10** in accordance with embodiments of the present disclosure is shown. The balance board **10** includes a platform **20** and one or more balance block(s) **40**. In use, a user stands on the balance board **10** with a single leg to train and to improve balance. Specifically, the platform **20** is sized and dimensioned to receive a single foot of a user as opposed to both feet of a user. Balance training on a single leg may translate to natural locomotion more readily than with two leg balance training. The balance board **10** may stimulate a user with both proprioceptively and mechanoreceptively simultaneously to activate more nerves in the foot and leg of the user. The dual stimulation of proprioceptive stimulation and mechanoreceptive stimulation may enhance somatosensory stimulation and may further benefit balance training. Proprioceptive stimulation is provided by micro-wobbles of the balance blocks **40**. Mechanoreceptive stimulation is provided by a mechanoreception stimulation surface **30** disposed on the platform **20**. The dual stimulation mechanisms may improve a rate of balance training.

[0036] With additional reference to FIG. **2**, the platform **20** has a first side or a top side **22**, a second side or a bottom side **24**, a first side end **26**, and a second side end **28**. The top side **22** of the platform **20** is sized and dimensioned to fit a single foot of a user thereon. The platform **20** may have a width in the range of 100 millimeters to 150 millimeters and may have length in the range of 300 millimeters to 380 millimeters. The platform **20** may be substantially rigid such that the platform **20** does not flex under the weight of the user. The user stands on the top side **22** of the platform **20** with a single foot unsecured to the platform **20**. For example, the balance board **10** may be used to improve the balance of a user during locomotion without a strap to secure the balance board **10** to the single foot of a user.

[0037] The top side **22** includes the mechanoreception stimulation surface **30**. The mechanoreception stimulation surface **30** has a plurality of protuberances **32** protruding upwardly from the top side **22**. When a user stands on the platform **20**, the protuberances **32** mechanoreceptively stimulate the foot of the user. The mechanoreception stimulation surface **30** may increase micro-circulation within the foot of the user. The protuberances **32** may mechanoreceptively stimulate the foot of the user and increase micro-circulation through texture, skin stretching, or pressure. For example, each protuberance **32** may have a peak and that acts as a pressure point on the foot of the user. In some embodiments, a valley is defined between adjacent protuberances **32** to receive the skin of the foot and stretch the skin. The smallest blood vessels of the circulatory system may have little to no smooth muscle surrounding the vessels to circulate blood. The mechanoreception stimulation surface **30** may assist circulation of blood through these smallest blood vessels. To maximize mechanoreceptive stimulation a user may stand on the mechanoreceptive stimulation surface **30** barefoot or with a thin sock or foot covering. The mechanoreception stimulation surface **30** may be made of a rubber ethylene vinylene acetate (RBEVA) material. The mechanoreception stimulation surface **30** may be adhered or otherwise bonded to the top side **22**. The mechanoreception stimulation surface **30** may cover only a portion of the top side **22** or may cover the entirety of the top side **22**. For example, a portion of the top side **22** may be exposed or uncovered by the mechanoreception stimulation surface **30**. The top side **22** may be exposed adjacent to the first side end **26** or the second side end **28**. In some embodiments, the top side **22** may be exposed adjacent to the first side end **26** and the second side end **28** as shown in FIG. **1**.

[0038] With particular reference to FIG. **3**, each protuberance **32** may have a pyramidal profile. In some embodiments, the protuberances **32** have a square base or a triangular base. In certain embodiments, the protuberances **32** are conical and have a circular base. The protuberances **32** may terminate in apex or may be frustopyramidal or frustoconical. The protuberances **32** may have a height in the range of 0.5 millimeters to 2 millimeters, e.g., 1.5 millimeters. The base of the protuberances **32** may have a width in the range of 0.5 millimeters to 4 millimeters, e.g., 2.5 millimeters. The base of the protuberances **32** may be in contact with one another or spaced apart in the range of 0.25 millimeters to 2 millimeters, e.g., 1 millimeter.

[0039] Referring to FIG. **4**, the bottom side **24** includes a securement surface **34** for attachment of the balance blocks **40** to the bottom side **24** of the platform **20**. The securement surface **34** extends along a length of the bottom side **24** between the first side end **26** and the second side end **28**. The securement surface **34** may be the first part or the second part of a two-part fastener, e.g., a hook and loop fastener. The securement surface **34** may cover the entirety of the bottom side **24** or may cover only a portion of the bottom side **24** as shown.

[0040] The balance blocks **40** attach to the bottom side **24** of the platform **20** and micro-wobble to proprioceptively stimulate the user. The balance blocks **40** may be made of a semi-rigid material that deforms under the weight of the user standing on the platform **20** of the balance board **10**. For example, the balance blocks **40** may be made of an open cell or closed cell foam material. Micro-wobbles may be generated under the weight of the user standing on the platform **20**. The micro-wobbles may be a result of muscle imbalances in the foot, ankle, knee, or hip of the user which may cause the balance blocks **40** to deform non-uniformly. The micro-wobbles of the balance blocks **40** may cause the user to adjust or compensate to maintain balance on the balance board **10** thereby training and improving the balance of the user. Each balance block **40** may deform omnidirectionally, 360-degrees, around the perimeter of the balance block **40**. The omnidirectional deformation of the balance blocks **40** allow for training balance in each plane or about each axis of movement. For example, deformation of the balance blocks **40** longitudinally along the length of the platform **20** towards the first side end **26** may train plantar flexion or dorsi-flexion and deformation of the balance blocks **40** laterally along the width of the platform **20** may train inversion or eversion of the foot and ankle.

[0041] With additional reference to FIGS. **5-8**. the balance blocks **40** include an attachment surface **42** that cooperates with the securement surface **34**. The attachment surface **42** may be the first part or the second part of the two-part fastener, e.g., the hook or the loop of a hook and loop fastener. The balance blocks **40** may secure to the bottom side **24** of the platform **20** at any point along the securement surface **34** between the first and second side ends **26**, **28**. The balance board **10** may have a plurality of configurations to adjust the difficulty of the balance training. For example, the balance board **10** may have a first or easy configuration, a second or moderate configuration, and a third or difficult configuration. In the easy configuration, a first balance block **40** and a second balance block **40** may be spaced apart and attached adjacent to the first side end **26** and the second side end **28** (FIG. **6**). In the moderate configuration, the first and second balance block **40** may be attached centrally or medially along securement surface **34** spaced apart from the first and second side ends **26**, **28**, and may be adjacent to or in contact with one another (FIG. **7**). In the difficult configuration, a single balance block **40** may be attached centrally or medially along the securement surface **34** (FIG. **8**). Adjusting the position of the balance blocks **40** may shift the balance point or equilibrium point of the balance board **10**.

[0042] With particular reference to FIGS. **9-12**, the balance blocks **40** may be formed in a variety sizes and shapes to further adjust the difficulty of balance training. The balance blocks **40** may have a rectangular or square profile or base. The balance blocks **40** have a height H in the range of 10 millimeter to 60 millimeters. The greater the height H of the balance blocks **40** the higher the center of gravity of the balance board **10** is positioned and, thus, the more difficult the balance training. In some embodiments, the balance blocks **40** have a semi-cylindrical profile. The semi-cylindrical profile may be a half a cylinder or may be less than half of a cylinder. For example, the semi-cylindrical profile may be taken along the diameter of a cylinder or may be taken along a chord of a cylinder. In such embodiments, the balance blocks **40** may roll in addition to or alternatively to micro-wobbling to increase the difficulty of the balance training. When a balance block **40** having a semi-cylindrical profile is attached to the platform **20** a central longitudinal axis of the balance block **40** may be parallel to a central longitudinal axis or the bottom side **24** of the platform **20**.

[0043] In certain embodiments, the balance board **10** may be configured to train and improve stability of the upper body. For example, a user may perform planks or push-ups on one or more balance boards **10** to train stability of the hands, wrist, elbow, and shoulder. In such embodiments, the platform **20** may be sized and dimensioned to fit a hand or both hands of user thereon.

[0044] With specific reference to FIG. **9**, a balance trainer kit **100** may include a platform **20** and a plurality of balance blocks **40***a*, **40***b*, **40***c* to compose the balance board **10** into a first or beginner composition, a second or intermediate composition, and a third or advanced composition. In the beginner composition (FIG. **10**), the balance blocks **40***a* are attached to the platform **20**. The balance blocks **40***a* may have the shortest height H of the plurality of balance blocks **40***a*, **40***b*, **40***c*. The balance blocks **40***a* may have a height Ha in the range of 10 millimeters to 30 millimeters, e.g., 25 millimeters. In the intermediate composition (FIG. **11**), the balance blocks **40***b* may be attached to the platform **20** and may have the tallest height H of the plurality of balance blocks **40***a*, **40***b*, **40***c*. The balance block **40***b* may have a height Hb in the range of 30 millimeters to 60 millimeters, e.g., 50 millimeters. In the advanced composition (FIG. **12**), the balance block **40***c* may be attached to the platform **20** and may have a semi-cylindrical profile. The balance block **40***c* may have a height Hc in the range of 20 millimeters to 50 millimeters, e.g., 35 millimeters.

[0045] The balance board **10** may be composed in any one of the beginner composition, intermediate composition, or advanced composition and may be configured into anyone of the easy configuration, the moderate configuration, or the difficult configuration. For example, the balance board **10** may be in the intermediate composition with the balance blocks **40***b* attached to the platform **20** and the easy configuration with the balance blocks **40***b* attached adjacent to the first side end **26** and the second side end **28**. In embodiments, the balance board **10** may be in the beginner composition and the difficult configuration with a single balance block **40***a* attached to platform **20** centrally along the securement surface **34**. In certain embodiments, the balance board **10** may be in the advanced composition and the moderate configuration with the balance blocks **40***c* attached to the platform **20** spaced apart from the first side end **26** and the second side end **28**.

[0046] With reference to FIG. **13**, a method **1000** of assembling a balance board in accordance with the present disclosure is described with reference to the balance board **10** of FIGS. **1-12**. Assembling the balance board **10** includes selecting a configuration and a composition for the balance board **10** (Step **1100**). The configuration of the balance board **10** may be one of the easy configuration, the moderate configuration, or the difficult configuration described above. The configuration of the balance board **10** refers to the number and the position of the balance blocks **40** attached to the bottom side **24** of the platform **20**. The composition of the balance board **10** may be one of the beginner composition, the intermediate composition, or the advanced composition described above. The composition of the balance board **10** refers to the type of balance block **40**, e.g., the balance block **40***a*, **40***b*, **40***c*, attached to the bottom side **24** of the platform **20**.

[0047] The balance blocks **40** are attached to the bottom side **24** of the platform **20** to assemble the balance board **10** in the selected configuration and composition (Step **1200**). The balance blocks **40** are attached through cooperation between the securement surface **34** and the attachment surface **42**. In embodiments, the securement surface **34** is the loop half of a hook and loop fastener and the attachment surface **42** is the hook half of the hook and loop fastener. In such embodiments, the balance blocks **40** are positioned along the securement surface **34** and pressed to attach the balance blocks **40** in the selected configuration with the selected composition.

[0048] The balance board **10** may be reconfigured or recomposed into another configuration or composition (Step **1300**). The other configuration or composition may be a different one of the easy configuration, the moderate configuration, or the difficult configuration or a different one of the beginner composition, the intermediate composition, or the advanced composition. For example, a balance board **10** in the easy configuration and the beginner composition may be reconfigured into the moderate configuration by removing the balance blocks **40***a* and reattaching the balance blocks **40***a* centrally or medially along securement surface **34** spaced apart from the first and second side ends **26**, **28**, and may be adjacent one another. The balance board **10** would then be in the moderated configuration and the beginner composition.

[0049] Although the method steps are described in a specific order, it should be understood that other steps may be performed in between described steps, described steps may be adjusted so that they occur at slightly different times, or the described steps may occur in any order unless otherwise specified.

[0050] While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Any combination of the above embodiments is also envisioned and is within the scope of the appended claims. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope of the claims appended hereto.

What is claimed:

1. A balance board comprising:

a platform having a top side and a bottom side opposite the top side, the platform sized and dimensioned to receive a single foot of a user thereon, the top side of the platform includes a mechanoreception stimulation surface configured to mechanoreceptively stimulate neurons within cutaneous skin of the user; and

a first balance block removably attached to the bottom side of the platform, the first balance block configured to proprioceptively stimulate neurons within muscles, tendons, or joints of the user.

2. The balance board according to claim **1**, wherein the top side is configured to receive the single foot of the user in an unsecured manner.

3. The balance board according to claim **1**, wherein the mechanoreception stimulation surface comprises a plurality of protuberances projecting away from the top side of the platform.

4. The balance board according to claim **1**, wherein the first balance block is configured to micro-wobble under the weight of a user to proprioceptively stimulate neurons of the user.

5. The balance block according to claim **4**, wherein the first balance block is configured to micro-wobble omnidirectionally around a perimeter thereof to improve balance of the user in each plane or axis of movement.

6. The balance board according to claim **1**, further comprising a second balance block removably attached to the bottom side of the platform, the second balance block sized and dimensioned similar to the first balance block.

7. The balance board according to claim **6**, wherein the balance board has:

a first configuration in which the first balance block is disposed adjacent to a first side end of the platform and the second balance block is disposed adjacent to a second side end of the platform opposite the first side end, the second balance block spaced apart from the first balance block; and

a second configuration in which the first balance block is spaced apart from the first side end and the second balance block is spaced apart from the second side end, the balance board configured to provide more proprioceptive stimulation when in the second configuration than when in the first configuration.

8. A balance board comprising:

a platform having a textured first side and a second side opposite the textured first side, the second side including a first part of a two-part fastener, the platform sized and dimensioned to receive a single foot of a user thereon, the textured first side configured to mechanoreceptively stimulate the user; and

a first balance block including a second part of the two-part fastener, the first balance block removably attached to the second side of the platform, the first balance block configured to proprioceptively stimulate the user.

9. The balance board according to claim **8**, wherein the textured first side comprises a plurality of protuberances projecting upwardly therefrom.

10. The balance board according to claim **8**, wherein the first balance block is configured to deform under the weight of the user to proprioceptively stimulate neurons of the user.

11. The balance board according to claim **10**, wherein the first balance block is configured to deform omnidirectionally around a perimeter thereof under the weight of the user to improve balance of the user in each plane or axis of movement.

12. The balance board according to claim **7**, wherein the balance board is configured to improve balance of the user with the single foot of the user unsecured to the platform.

13. A balance trainer kit comprising:

a balance board according to claim **8**; and

a plurality of balance blocks attachable to the platform such that the balance board is configurable into a first configuration, a second configuration, and a third configuration and is composable into a first composition and a second composition.

14. A balance board comprising:

a platform having a top side and a bottom side opposite the top side, the top side including mechanoreception stimulation surface, the bottom side including a securement surface extending along a length of the bottom side between a first side end of the platform and a second side end opposite the first side end; and

a first balance block removably attached to the bottom side of the platform and positionable between the first side end and the second side end, the first balance block including an attachment surface mateable with the securement surface to attach the first balance block to the platform.

15. The balance board according to claim **14**, wherein the first balance block has a rectangular profile.

16. The balance board according to claim **15**, wherein the first balance block has a height in the range of 10 millimeters to 60 millimeters.

17. The balance board according to claim **14**, wherein the first balance block has semi-cylindrical profile.

18. The balance board according to claim **14**, wherein the securement surface is a first part of a two-part fastener and the attachment surface is a second part of the two-part fastener.

19. The balance board according to claim **14**, comprising a second balance block removably attached to the bottom side of the platform and positionable between the first side end and the second side end.

20. The balance board according to claim **19**, wherein the balance board has a first configuration in which the first balance block is disposed adjacent to the first side end and the second balance block is disposed adjacent to the second side end and spaced apart from the first balance block and a second configuration in which the first balance block is spaced apart from the first side end and the second balance block is spaced apart from the second side end.

21. A method of assembling a balance board comprising:

selecting a configuration and a composition for the balance board from one of a first configuration, a second configuration, or a third configuration and a first composition, a second composition, or a third composition; and

attaching balance blocks to a platform of the balance board such that the balance board is in the selected configuration and has the selected composition.

22. The method according to claim **21**, comprising reconfiguring the balance board into a different configuration of the first configuration, the second configuration, or the third configuration.

23. The method according to claim **21**, comprising recomposing the balance board into a different composition of the first composition, the second composition, or the third composition.

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